

1.6 Rules of Inference

Proofs in mathematics are valid arguments that establish the truth of mathematical statements. By an argument, we mean a sequence of statements that end with a conclusion. By valid, we mean that the conclusion, or final statement of the argument, must follow from the truth of the preceding statements, or premises, of the argument. That is, an argument is valid if and only if it is impossible for all the premises to be true and the conclusion to be false. To deduce new statements from statements we already have, we use rules of inference which are templates for constructing valid arguments. Rules of inference are our basic tools for establishing the truth of statements.

1.6.2 Valid Arguments in Propositional Logic

Consider the following argument involving propositions (which, by definition, is a sequence of propositions):

"If you have a current password, then you can log onto the network."

"You have a current password."

Therefore,

"You can log onto the network."

a) Find out whether this is a valid argument

Form of the argument

Use p to represent "You have a current password" and q to represent "You can log onto the network." Then, the argument has the form

$p \rightarrow q$
p
$\therefore q$

By Ex-1.3, Prob (12(c)), we know that when p and q are propositional variables, the statement $(p \rightarrow q) \wedge p \rightarrow q$ is a tautology. In particular, when both $p \rightarrow q$ and p are

true, we know that q must also be true. We say this form of argument is valid because whenever all its premises are true, the conclusion must also be true.